EXPERIMENT NO 1:

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim:

To implement and demonstrate the **FIND-S Algorithm** for finding the most specific hypothesis that is consistent with the given set of positive training examples.

Procedure:

1. Define the training dataset in the program.
2. Initialize the hypothesis using the first positive training example.
3. Compare the hypothesis with each positive example in the dataset.
4. If an attribute value differs, replace it with '?' to generalize the hypothesis.
5. Ignore all negative training examples.
6. Continue until all training examples are processed.
7. Display the final most specific hypothesis.

Program:

```
# FIND-S Algorithm

# Training Dataset

data = [

   ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

   ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],

   ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],

   ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

]

# Number of attributes

num_attributes = len(data[0]) - 1

# Initialize hypothesis

hypothesis = None

# Find the first positive example

for row in data:
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    if row[-1] == "Yes":
        hypothesis = row[:-1]
        break
# Apply FIND-S Algorithm
for row in data:
    if row[-1] == "Yes":
        for i in range(num_attributes):
            if hypothesis[i] != row[i]:
                hypothesis[i] = '?'
# Display the results
print("Training Data:\n")
for row in data:
    print(row)
print("\nMost Specific Hypothesis:")
print(hypothesis)
```

OUTPUT:

```
Training Data:

['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']
['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']
['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No']
['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result:

The FIND-S algorithm was successfully implemented and executed. The most specific hypothesis obtained from the given training data is: